

GUILLAUME STEIMER



Strasbourg, France



guillaume.steimer.pro@gmail.com



Web page



Linkedin

EDUCATION

- 2021 – 2025 **PhD in Applied Mathematics** **University of Strasbourg**
Model order reduction method for Hamiltonian dynamics using deep learning
Development of numerical models for the simulation of complex physical systems including wave, fluid and plasma dynamics, design of physics-based non-linear model order reduction algorithms using machine learning, Python implementation and optimization, preservation of physical invariants, model validation and performance analysis.
- 2019 – 2021 **Master in Scientific Computation and Mathematics of Innovation (CSMI)** **University of Strasbourg**
Advanced training in numerical methods for ODEs/PDEs, modeling, simulation, machine learning, high-performance computing, optimization, scientific computing, graphs, signal processing, compilation, uncertainties quantification, networks.
Graduated with honors (très bien)
- 2016 – 2019 **Bachelor in Mathematics** **University of Strasbourg**
2 years of BSc (DEUG) in Advanced Mathematics and Physics (MPA), 3rd year in Applied Mathematics.
Graduated with honors (très bien)

PUBLICATION & PREPRINT

Reduced particle-in-cell method for the Vlasov–Poisson system using autoencoder and Hamiltonian neural networks

R. Côte , E. Franck , L. Navoret , G. Steimer, V. Vigon.

doi:10.1007/s10444-026-10364-x

Adv. Comput. Math., 2026

Hamiltonian reduction using a convolutional auto-encoder coupled to an Hamiltonian neural network

R. Côte , E. Franck , L. Navoret , G. Steimer, V. Vigon.

10.4208/cicp.OA-2023-0300

Commun. in Comput. Phys., 2025

Hyperbolic reduced model for Vlasov-Poisson equation with Fokker-Planck collision

E. Franck, I. Lannabi, Y. Nasser, L. Navoret, G. Parasiliti Rantone, G. Steimer.

10.1051/proc/202477213

ESAIM Proc. Surveys, 2024

EXPERIENCE IN NUMERICAL SIMULATION AND MODELING

- 2/2021 – 8/2021 **Master's internship (MSc Year 2): Model order reduction in plasma dynamics for nuclear fusion applications** **INRIA (team TONUS), IRMA, Strasbourg**
Development of a reduced-order model for the simulation of a complex physical system: implementation of a Particle-In-Cell (PIC) method in Python, design of machine-learning-based model reduction algorithms, validation and verification of results.
- 10/2020 – 1/2021 **Master's project (MSc Year 2): Learning physical dynamics using neural networks** **Team MOCO, IRMA, Strasbourg**
Implementation in Python of machine learning models capable of learning the dynamics of conservative systems, stability analysis and preservation of physical invariants.
- 3/2020 – 8/2020 **Master's project and internship (MSc Year 1): Modeling and simulation for tsunami prediction** **Team MOCO, IRMA, Strasbourg**
Development of a 1D C++ simulation code based on fluid dynamics equations, integration of real topographic data, tsunami arrival time prediction using neural networks, numerical performance analysis.

INTERNATIONAL CONFERENCES

- June 2025 **4th International Nonlinear Dynamics Conference (NODYCON 2025)** **New-York, USA**
Reduced Particle in Cell method for the Vlasov-Poisson system using auto-encoder and Hamiltonian neural networks
E. Franck, L. Navoret, G. Steimer, V. Vigon

February 2025	3rd IACM Digital Twins in Engineering Conference & 1st ECCOMAS Artificial Intelligence and Computational Methods in Applied Science (DTE & AICOMAS) Reduced Particle in Cell method for the Vlasov-Poisson system using auto-encoder and Hamiltonian neural networks E. Franck, L. Navoret, G. Steimer, V. Vigon	Paris, France
May 2023	Math 2 Product (M2P) : Emerging Technologies in Computational Science for Industry, Sustainability and Innovation Reduced order modeling using auto-encoder and Hamiltonian neural networks E. Franck, L. Navoret, G. Steimer, V. Vigon	Taormina, Italy
November 2022	Numerical Methods for the Kinetic Equations of Plasma Physics (NumKin) Data driven reduced modeling of the Vlasov-Poisson equation E. Franck, L. Navoret, G. Steimer, V. Vigon	Garching, Germany
September 2022	Model Reduction and Surrogate Modeling (MORE) Data driven reduced modeling of the Vlasov-Poisson equation E. Franck, L. Navoret, G. Steimer, V. Vigon	Berlin, Germany
June 2022	8th European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS) Data driven reduced modeling of the Vlasov-Poisson equation E. Franck, N. Crouseilles, L. Navoret, G. Steimer, V. Vigon	Oslo, Norway

TEACHING

2024-2025	Scientific Computing (BSc 2nd year in Mathematics) Fundamentals of numerical analysis, applications in Python.	University of Strasbourg
2021-2025	Computer Science (BSc 2nd year in Mathematics) Basics of object-oriented programming in Python.	University of Strasbourg
2021-2024	Computer Science (BSc 3rd year in applied mathematics) Object-oriented programming in C++.	University of Strasbourg

TECHNICAL SKILLS

Simulation & Modeling: Numerical methods for ODEs/PDEs, scientific machine learning, physical modeling, model order reduction, numerical analysis, high-performance computing, signal processing, statistical analysis, optimization, uncertainty quantification, general physics background.

Software Development: Python, Tensorflow, Pytorch, Scikitlearn, Keras, C, C++, Git, Linux, working knowledge of bash, MATLAB, OpenMP, MPI, CUDA, Openturns, Pandas, SQL.

Languages: French (native), English (fluent).